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Compute Once, Distribute Many: Efficient Feature Pyramid Refinement for Ultra-Lightweight Detectors

Haoming Guo, Wei Li, Xiaoyan Yang, Yumin Chen, Yuping Song

***Abstract*—The YOLO series has achieved remarkable progress in real-time object detection, yet nano-scale variants remain constrained by single-round feature pyramid topologies where the global correlation prior is distributed only once and extreme pyramid levels exchange information indirectly. In this paper, we propose YOLO-Prism, an efficient nano-scale detector built upon YOLOv13 that addresses this single-round limitation through asymmetric multi-round refinement. We propose a Multi-Round Feature Pyramid Network (MR-FPN) that exploits the decoupling between YOLOv13's precomputed correlation prior and its lightweight gated distribution mechanism, appending a secondary top-down refinement pass at only $\sim$6% parameter overhead under a "compute once, distribute many" paradigm. Subsequently, as a near-zero-cost complement, we selectively integrate DCT-based Multi-Spectral Channel Attention (MSCA) at gradient-identified critical layers, contributing an additional gain at merely 0.01M parameters. Extensive experiments on MS COCO, PASCAL VOC, and VisDrone benchmarks demonstrate that YOLO-Prism achieves 41.66% $AP_{50:95}$ on COCO val2017, surpassing the strong YOLOv13n baseline by 0.62 points under identical training conditions and outperforming YOLOv12n by 1.10 points, at 2.63M parameters and 7.1 GFLOPs. On YOLOv11n, whose neck lacks such a prior, the same asymmetric topology yields 0.50 $AP_{50:95}$ at 3.8% overhead, confirming that the benefit of multi-round refinement is independent of any specific gating mechanism. This lightweight algorithm can be applied to intelligent video measurement, industrial edge detection sensors, and IoT monitoring.**

***Index Terms*—Object detection, feature pyramid network, multi-round fusion, frequency-domain channel attention.**

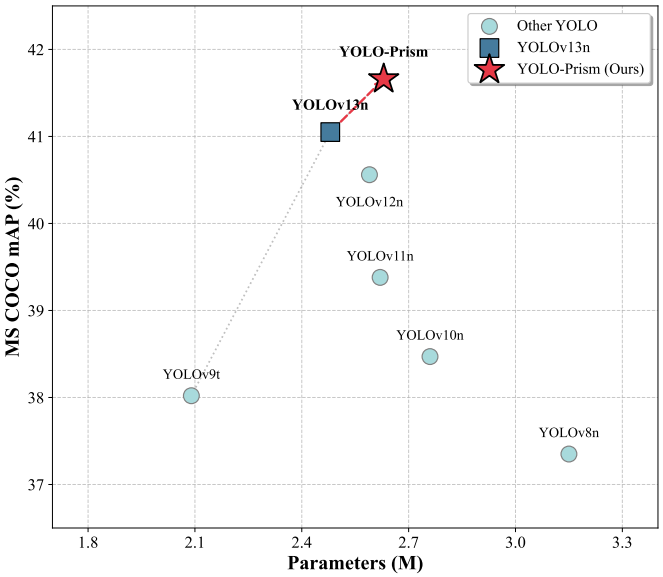


Fig. 1. MS COCO mAP versus parameters on COCO val2017. YOLO-Prism achieves the highest accuracy among all compared nano-scale detectors with only a modest parameter increase over the baseline.

I. Introduction

OBJECT detection underpins autonomous driving [1], robotics, and video surveillance [2]. The field has evolved from two-stage detectors [3] to efficient single-stage frameworks such as YOLO [4], [5] and Transformer-based detectors such as RT-DETRv2 [6]. Although recent models including YOLOv6 [7], YOLOv9 [8], and YOLOv10 [9] continue pushing the accuracy–speed Pareto frontier, these advances are largely driven by increased model capacity. When constrained to the ultra-lightweight "Nano" regime of $\leq$3M parameters, semantic representation degrades markedly [10]: narrow channels compress each pyramid level into a low-dimensional space, limiting discriminative expressiveness, while shallow fusion topologies restrict cross-scale information flow, leaving small objects under-represented at deeper layers.

YOLOv13 [11] currently represents the state of the art in the nano regime. Its Hypergraph-based Adaptive Correlation Enhancement (HyperACE) module produces a *global correlation prior* $\mathcal{H}$—a hypergraph-encoded representation of inter-scale correlations—which is distributed via Full-Pipeline Aggregation-and-Distribution (FullPAD) modules to capture richer inter-scale dependencies than conventional FPNs [12], [13]. However, its pyramid is confined to a single top-down and bottom-up round: the expensive prior $\mathcal{H}$ is distributed only once, and extreme levels $\mathbf{P}_3$ and $\mathbf{P}_5$ exchange information only indirectly. Existing multi-round FPNs such as BiFPN [14] address routing limitations by stacking bi-directional rounds. However, their approach of duplicating entire fusion modules

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inflates parameters prohibitively for nano-scale models [15], [16]. A BiFPN-style second round on YOLOv13n would add approximately 0.8M parameters (32% overhead), violating the nano parameter budget. A separate limitation arises at the micro level within each fusion block, where standard spatial pooling discards higher-frequency information critical for fine-grained discrimination.

To address these challenges, we observe a structural opportunity inherent to gate-based prior-distribution architectures such as YOLOv13: the global correlation prior $\mathcal{H}$ and its distribution mechanism, FullPAD, are *decoupled*. Each FullPAD gate is a single learnable scalar per level, initialized to zero, so new distribution points can be instantiated at negligible cost without replicating the expensive HyperACE computation. Exploiting this property, we propose our primary contribution, **Multi-Round Feature Pyramid Network (MR-FPN)**, which appends a secondary top-down refinement pass through newly introduced FullPAD gates that reuse the precomputed $\mathcal{H}$. Unlike conventional multi-round designs that duplicate complete fusion modules at every round, MR-FPN follows a "compute once, distribute many" paradigm where each additional round adds only a compact refinement block and one scalar gate per level. To complement this macro-architecture enhancement, we further integrate DCT-based multi-spectral channel attention (MSCA) [17] into YOLOv13's bottlenecks as a drop-in replacement for standard spatial pooling. Crucially, MSCA contributes 0.14 $AP_{50:95}$ while adding merely 0.01M parameters and zero additional FLOPs, yielding a parameter efficiency roughly an order of magnitude beyond conventional attention mechanisms. Combining both components, we present **YOLO-Prism** as our primary model, featuring MR-FPN and selectively placed MSCA.

As shown in Fig. 1, YOLO-Prism achieves 41.66% $AP_{50:95}$ on COCO val2017, surpassing the YOLOv13n baseline by 0.62 points under identical training conditions with only a $\sim$6% parameter increase to 2.63M and competitive FLOPs of 7.1G. This gain, obtained through targeted architectural refinement of the feature pyramid alone, represents a meaningful advance on an already highly optimized baseline: YOLOv13n is the strongest nano-scale detector among those compared (Table II), and the 0.62-point improvement exceeds $20\times$ the baseline's inter-seed standard deviation (± 0.03). The main contributions of this work are summarized as follows:

- We propose MR-FPN, an asymmetric multi-round feature pyramid that reuses a precomputed global correlation prior across fusion rounds at near-zero marginal cost. By exploiting the decoupling between expensive prior computation and lightweight gated distribution, each additional round adds only a compact refinement block and one scalar gate per level, contributing 0.58 $AP_{50:95}$ on YOLOv13n. The same asymmetric topology transfers to YOLOv11n, yielding 0.50 $AP_{50:95}$ despite its conventional FPN-PAN neck, confirming that lightweight multi-round refinement is a general principle rather than a model-specific trick.
- As a near-zero-cost complement, we integrate DCT-based multi-spectral channel attention (MSCA) into select bottleneck layers, adding 0.14 $AP_{50:95}$ at merely 0.01M parameters. Gradient analysis of the combined topology reveals a structural sparsity pattern, with 9 of 14 attention positions collapsing to identity, from which we derive a principled selective placement strategy.
- YOLO-Prism achieves 41.66% $AP_{50:95}$ on COCO val2017, an improvement of 0.62 points over YOLOv13n, surpassing all compared nano-scale detectors at 2.63M parameters and 7.1 GFLOPs. Gains on VOC 2007 and YOLOv11n further support the generality of the approach, and the VisDrone results provide directionally consistent, evidence for small-object surveillance.

II. Related Work

A. Real-Time Object Detection

Real-time object detection requires balancing precision and latency. The YOLO series [4], [5] pioneered the single-stage paradigm, transforming detection into an end-to-end regression problem. Subsequent iterations have continuously pushed the Pareto frontier: YOLOv5 [18] and YOLOv6 [7] introduced hardware-friendly designs; YOLOv9 [8] and YOLOv10 [9] explored programmable gradients and NMS-free architectures; YOLO-World [19] broadened open-vocabulary paradigms; and real-time Transformers such as RT-DETRv2 [6] achieved state-of-the-art results. Constrained environments further motivate compact designs for tiny objects [10], [20] and efficient event-based networks [21]. Beyond general-purpose detection, domain-specific lightweight YOLO variants have been developed for precision agriculture [22] and dense large-scene detection [23]. Specialized network architectures have been developed to handle diverse and challenging detection scenarios [24]–[27]. Small-object detection has been improved through advanced attention and multi-scale fusion mechanisms [28]–[31]. Multimodal fusion and efficient 3D detection have been explored for intelligent transportation systems [32]–[35]. Efficient representation and control techniques have been proposed for point-cloud and wearable sensing applications [36]–[38]. Machine learning techniques have also been applied to healthcare imaging and biomechanical assessment [39]–[41].

These diverse applications underscore a common need for accurate yet efficient visual perception under stringent resource constraints. Despite these advances, however, achieving rich semantic feature alignment in ultra-lightweight nano-scale models remains an open challenge. This motivates our work, which builds upon YOLOv13 [11] and proposes a multi-round fusion strategy to enrich cross-scale interaction under tight parameter budgets.

B. Cross-Scale Feature Aggregation

Robust feature pyramids are essential for multi-scale object detection. Since FPN [12], multi-scale aggregation has become standard in modern detectors. Subsequent designs have sought to enhance information flow: PANet [13] added bottom-up augmentation; EfficientDet [14] proposed BiFPN, which stacks bi-directional fusion rounds; and recent work has explored hierarchical residual-like connections [42], unified

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TABLE I
SUMMARY OF RECENT SURVEY AND REVIEW PAPERS RELEVANT TO LIGHTWEIGHT OBJECT DETECTION.

Survey	Year	Scope	Key Insight Relevant to This Work
Liu *et al.* [2]	2020	Generic object detection: frameworks, features, proposals, context, training, evaluation (300+ papers)	Nano-scale detectors face a fundamental accuracy–efficiency tension; architectural innovation beyond capacity scaling is needed
Tian *et al.* [52]	2025	YOLO evolution from v1 to v13: anchors, multi-scale prediction, attention, lightweight design, anchor-free architectures	Multi-scale prediction, attention modules, and lightweight design are three core drivers of YOLO progress
Altaie *et al.* [53]	2025	Lightweight YOLO optimization for edge/IoT: structural modules (ShuffleNet, MobileNet, GhostNet) and model compression (distillation, quantization, pruning)	Hybrid strategies (structural modification + compression) yield the best accuracy–efficiency balance for resource-constrained deployment
Wang *et al.* [54]	2025	Systematic literature review of 103 lightweight YOLO papers (2016–2025): pruning, quantization, distillation, lightweight architectures; deployment on Jetson, RPi, MCUs, FPGAs	Hardware-aware co-design and standardized evaluation protocols are identified as open challenges for edge deployment

scale and spatial awareness [43], gather-and-distribute mechanisms [16], and neural architecture search for automatic FPN topology discovery [44]. However, BiFPN's repetitive stacking incurs substantial overhead, limiting its applicability to nano-scale models.

To overcome this, iterative and recursive pyramid strategies have been explored. DetectoRS [15] feeds FPN outputs back into the backbone for iterative refinement at substantially increased computation. AFPN [45] progressively bridges non-adjacent pyramid levels to mitigate semantic gaps. RT-DETRv3 [46] demonstrates the benefit of multi-round feature interaction through hierarchical dense positive supervision. Common to these approaches, however, is the replication of complete fusion blocks at each iteration, which leads to prohibitive parameter growth at nano scale. For example, applying a BiFPN-style second round to YOLOv13n would duplicate all fusion blocks and add approximately 0.8M parameters—a 32% increase that alone exceeds the entire parameter budget of our MR-FPN (0.15M) by more than fivefold. By contrast, our MR-FPN repurposes YOLOv13's native gating modules to distribute the global correlation prior across multiple rounds without duplicating expensive computations: the prior $\mathcal{H}$ is computed once by HyperACE and distributed to all rounds through lightweight scalar gates, so each additional round introduces only a compact refinement block and a single learnable scalar per level. This decoupling of per-round cost from prior computation complexity yields a 5.3$\times$ parameter efficiency advantage over conventional multi-round stacking.

C. Channel Attention and Frequency Domain Representation

Attention mechanisms selectively emphasize informative features. Squeeze-and-Excitation (SE) [47] established channel attention using Global Average Pooling (GAP) to compress spatial cues. CBAM [48] extended this by jointly modeling spatial and channel attention; Coordinate Attention [49] embedded positional information; and ECA-Net [50] further simplified the design with lightweight local cross-channel interactions.

GAP, however, mathematically retains only the DC component, discarding texture and boundary information encoded in higher frequencies. FcaNet [17] addressed this by generalizing channel attention into the frequency domain, replacing GAP with a multi-spectral 2D Discrete Cosine Transform (DCT) encoding. Building on this paradigm, recent work [51] has modulated dynamic attention via frequency properties. These advances demonstrate that multi-spectral representations offer fundamentally richer feature compression than single-frequency GAP. Yet existing frequency-domain attention mechanisms have been designed and validated primarily on medium-to-large models; their behavior in the ultra-narrow channel regime ($C \leq 64$), where the dual-FC bottleneck degenerates to a rank-1 mapping, remains unexplored. Motivated by this gap, our MSCA-Bottleneck adapts DCT-based channel attention into YOLOv13's depth-wise separable convolution blocks specifically for the nano regime. As we show in Section III-C, the extreme bottleneck ratio acts not as a limitation but as an implicit structural regularizer, contributing meaningful gains at negligible computational cost.

Several recent surveys provide broader context for the developments discussed above. Table I summarizes representative survey and review papers spanning generic object detection [2], the architectural evolution of YOLO [52], and systematic reviews of lightweight YOLO optimization for edge deployment [53], [54].

III. APPROACH

This section presents the core architecture of YOLO-Prism. Fig. 2 illustrates the complete pipeline. Before the mathematical details, we outline the design workflow in three stages: (i) HyperACE computes the global correlation prior $\mathcal{H}$ from the backbone features $\mathbf{P}_3$–$\mathbf{P}_5$ (Section III-A); (ii) MR-FPN distributes $\mathcal{H}$ across two fusion rounds, appending a lightweight top-down-only second round that reuses the same prior at near-zero marginal cost (Section III-B); (iii) MSCA recalibrates channel features at selected bottleneck layers in the frequency domain (Section III-C). Their individual and combined contributions are evaluated in Section IV-C, and the asymmetric multi-round design transfers to detectors without gated priors (Section IV-D).

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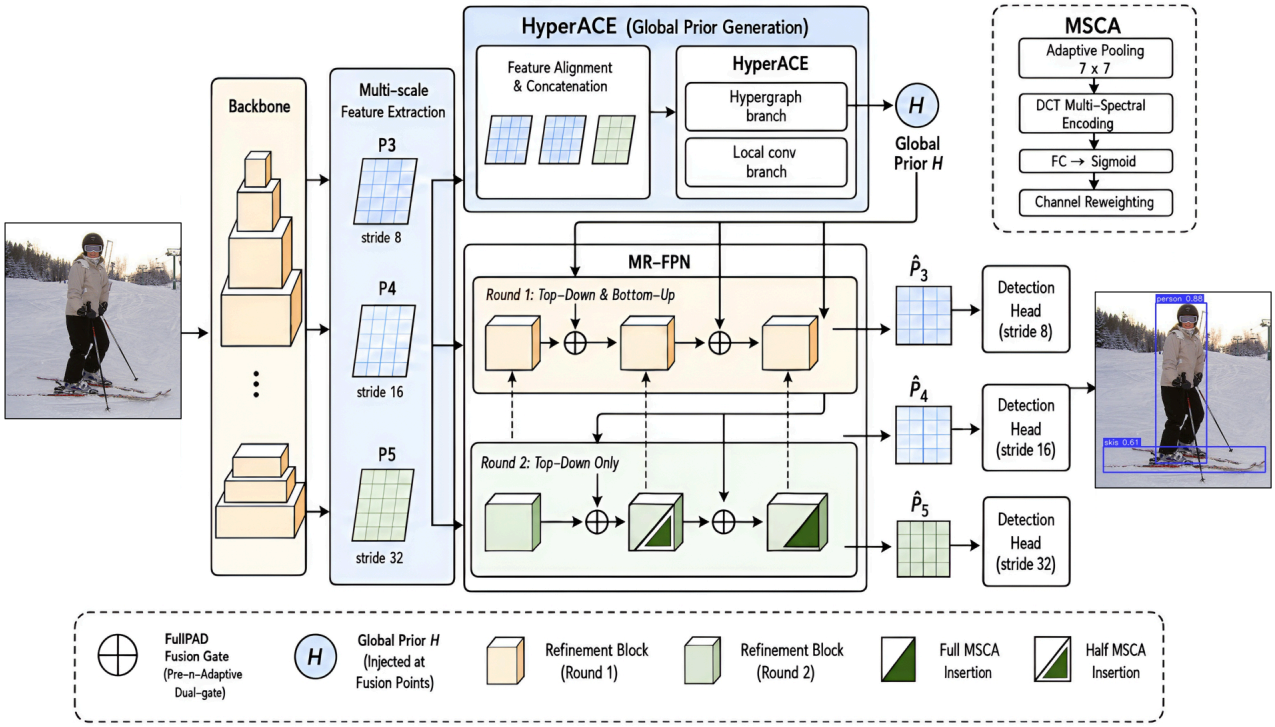


Fig. 2. Overall architecture of YOLO-Prism. The backbone extracts multi-scale features $\mathbf{P}_3$–$\mathbf{P}_5$, which are processed by HyperACE to produce a global correlation prior $\mathcal{H}$. MR-FPN distributes $\mathcal{H}$ through two rounds: Round 1 (orange) performs a full top-down and bottom-up cycle, while Round 2 (green) appends a lightweight top-down-only refinement reusing the same $\mathcal{H}$ via additional FullPAD gates. Selectively placed MSCA modules (dark green) provide frequency-domain channel recalibration at gradient-identified critical layers.

A. Preliminary

YOLOv13 [11] introduces two signature components that distinguish it from preceding YOLO variants.

Hypergraph-based Adaptive Correlation Enhancement (HyperACE). Given multi-scale backbone features $\{\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5\}$, HyperACE first aligns and concatenates them into a unified spatial representation, then models inter-scale correlations through two parallel branches: a *hypergraph convolution branch* that captures global high-order dependencies by constructing a hypergraph incidence matrix over channel groups, and a *local convolution branch* that preserves fine-grained spatial patterns via standard convolutions. The two branches are fused to produce a correlation-enhanced feature:

$$\mathcal{H} = f_{\text{HyperACE}}(\mathbf{P}_3,\ \mathbf{P}_4,\ \mathbf{P}_5)\,. \tag{1}$$

Because hypergraph edges can connect arbitrary subsets of nodes, this formulation captures *beyond-pairwise* channel interactions that standard graph convolutions cannot represent, endowing $\mathcal{H}$ with richer relational semantics than concatenation-based priors. The resulting $\mathcal{H}$ serves as a global prior distributed to individual pyramid levels.

Full-Pipeline Aggregation-and-Distribution (FullPAD). FullPAD distributes $\mathcal{H}$ through a gated residual fusion module called `FullPAD_Tunnel`:

$$\mathbf{Z}_l = \mathbf{P}_l + \gamma_l \cdot \mathcal{H}_l, \quad l \in \{3,4,5\}, \tag{2}$$

where $\mathbf{P}_l$ denotes the local feature at pyramid level l, $\mathcal{H}_l$ is the spatially aligned global prior, and γ_l is a learnable scalar gate initialized to 0, ensuring progressive introduction of the enhanced feature. This gated mechanism recurs throughout both the top-down (TD) and bottom-up (BU) pathways, forming a single-round TD–BU topology.

While HyperACE captures rich cross-scale correlations and FullPAD distributes them reliably, a key limitation remains: confining the computationally expensive global correlation $\mathcal{H}$ to a single fusion round under-utilizes its representational capacity. MR-FPN is designed to address this limitation by enabling multi-round distribution of $\mathcal{H}$ through the existing FullPAD infrastructure.

B. Multi-Round Feature Pyramid Network (MR-FPN)

Single-round pyramids limit $\mathbf{P}_3$–$\mathbf{P}_5$ interaction, while multi-round alternatives such as BiFPN duplicate complete fusion modules at prohibitive cost (Section II). MR-FPN achieves multi-round fusion by repurposing the *existing* FullPAD infrastructure: the global prior $\mathcal{H}$ is computed once and distributed to additional rounds through new FullPAD gates, so that each round adds only a compact `DSC3k2` refinement block and one scalar gate per level. Let $\mathcal{R}(\cdot\,; n)$ denote the `DSC3k2` refinement block with depth n; we instantiate MR-FPN with $T{=}2$ rounds, yielding three successive phases (Fig. 3), formalized in Algorithm 1.

In **Round 1 Top-Down**, the FullPAD-enhanced $\mathbf{P}_5$ is progressively upsampled, concatenated with the corresponding

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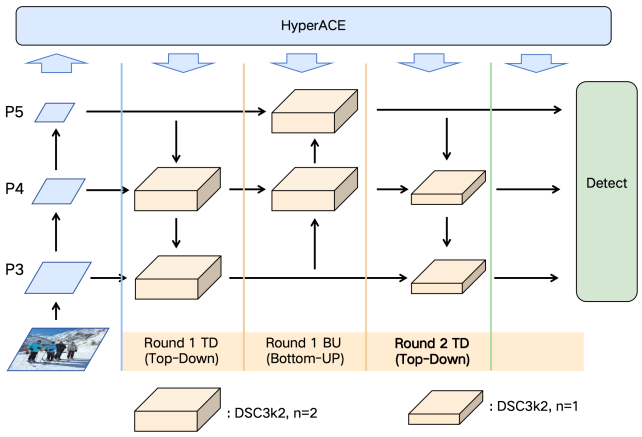


Fig. 3. Topology of the proposed MR-FPN. Round 1 performs a complete top-down and bottom-up cycle using `DSC3k2` blocks (n=2). Round 2 appends a lightweight top-down-only refinement using minimal `DSC3k2` blocks (n=1). The blue arrows from HyperACE indicate the FullPAD-gated distribution of the global feature $\mathcal{H}$ into each fusion node across all rounds.

backbone features, refined by $\mathcal{R}(\cdot\,;2)$, and gated with FullPAD:

$$\mathbf{P}_l^{(1,\mathrm{td})} = \mathcal{R}\big(\mathrm{Cat}(\uparrow\mathbf{P}_{l+1}^{(1,\mathrm{td})},\ \mathbf{P}_l)\,;\ 2\big) + \gamma_l^{(1,\mathrm{td})}\cdot\mathcal{H}_l, \quad l=4,3, \tag{3}$$

where $\uparrow$ denotes nearest-neighbor $2\times$ upsampling, $\mathrm{Cat}(\cdot)$ is channel-wise concatenation, and $\gamma_l^{(1,\mathrm{td})}$ is the FullPAD gate. In **Round 1 Bottom-Up**, the R1-TD features are propagated upward through strided convolutions and refined symmetrically, completing a full bidirectional interaction:

$$\mathbf{P}_l^{(1,\mathrm{bu})} = \mathcal{R}\big(\mathrm{Cat}(\downarrow\mathbf{P}_{l-1}^{(1,\mathrm{td})},\ \mathbf{P}_l^{(1,\mathrm{td})})\,;\ 2\big) + \gamma_l^{(1,\mathrm{bu})}\cdot\mathcal{H}_l, \quad l=4,5, \tag{4}$$

where $\downarrow$ denotes $2\times$ strided convolution. At this point, $\{\mathbf{P}_3^{(1,\mathrm{td})}, \mathbf{P}_4^{(1,\mathrm{bu})}, \mathbf{P}_5^{(1,\mathrm{bu})}\}$ form the Round 1 output.

In **Round 2 Top-Down**, the Round 1 output is refined by a second top-down pass with compact $\mathcal{R}(\cdot\,;1)$ blocks. Since $\mathbf{P}_3$ does not participate in Round 1 BU, we denote each level's Round 1 output as $\tilde{\mathbf{P}}_l$: $\tilde{\mathbf{P}}_5 = \mathbf{P}_5^{(1,\mathrm{bu})}$, $\tilde{\mathbf{P}}_4 = \mathbf{P}_4^{(1,\mathrm{bu})}$, and $\tilde{\mathbf{P}}_3 = \mathbf{P}_3^{(1,\mathrm{td})}$. The Round 2 refinement then proceeds as:

$$\mathbf{P}_l^{(2,\mathrm{td})} = \mathcal{R}\big(\mathrm{Cat}(\uparrow\tilde{\mathbf{P}}_{l+1}^{\star},\ \tilde{\mathbf{P}}_l)\,;\ 1\big) + \gamma_l^{(2,\mathrm{td})}\cdot\mathcal{H}_l, \quad l=4,3, \tag{5}$$

where $\tilde{\mathbf{P}}_{l+1}^{\star}$ denotes the upstream source: $\tilde{\mathbf{P}}_5$ for l=4, and the freshly computed $\mathbf{P}_4^{(2,\mathrm{td})}$ for l=3. The detection heads receive $\{\mathbf{P}_3^{(2,\mathrm{td})}, \mathbf{P}_4^{(2,\mathrm{td})}, \tilde{\mathbf{P}}_5\}$, where the shallow levels have undergone two rounds of top-down semantic injection from $\mathcal{H}$. The key efficiency insight is that $\mathcal{H}$ is computed once by Eq. (1) and reused across all phases, keeping the marginal cost of each additional round negligible.

Two design principles shape MR-FPN's Round 2, independent of the underlying neck architecture. First, each Round 2 node adopts a *minimal per-node footprint*—one compact `DSC3k2` block (n=1) per level, plus, on gated architectures, a single scalar FullPAD gate—keeping the marginal cost negligible regardless of the number of rounds. Second, Round 2 performs only a top-down pass rather than a full

Algorithm 1: MR-FPN Forward Pass

Input: Backbone features $\{\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5\}$, global prior $\mathcal{H}$

Output: Detection-head inputs $\{\hat{\mathbf{P}}_3, \hat{\mathbf{P}}_4, \hat{\mathbf{P}}_5\}$

`// Round 1: Top-Down (depth n=2)`

$\mathbf{P}_5^{(1)} \leftarrow \mathbf{P}_5$;

for $l = 4, 3$ **do**

  $\mathbf{P}_l^{(1,\mathrm{td})} \leftarrow \mathcal{R}\big(\mathrm{Cat}(\uparrow\mathbf{P}_{l+1}^{(1)},\ \mathbf{P}_l)\,;\ 2\big) + \gamma_l^{(1,\mathrm{td})}\cdot\mathcal{H}_l$;

`// Round 1: Bottom-Up (depth n=2)`

for $l = 4, 5$ **do**

  $\mathbf{P}_l^{(1,\mathrm{bu})} \leftarrow \mathcal{R}\big(\mathrm{Cat}(\downarrow \mathbf{P}_{l-1}^{(1,\mathrm{td})},\ \mathbf{P}_l^{(1,\mathrm{td})})\,;\ 2\big) + \gamma_l^{(1,\mathrm{bu})}\cdot\mathcal{H}_l$;

`// Collect Round 1 outputs:` $\tilde{\mathbf{P}}_3{=}\mathbf{P}_3^{(1,\mathrm{td})}$, $\tilde{\mathbf{P}}_4{=}\mathbf{P}_4^{(1,\mathrm{bu})}$, $\tilde{\mathbf{P}}_5{=}\mathbf{P}_5^{(1,\mathrm{bu})}$

`// Round 2: Top-Down only (depth n=1)`

$\mathbf{P}_4^{(2,\mathrm{td})} \leftarrow \mathcal{R}\big(\mathrm{Cat}(\uparrow\tilde{\mathbf{P}}_5,\ \tilde{\mathbf{P}}_4)\,;\ 1\big) + \gamma_4^{(2,\mathrm{td})}\cdot\mathcal{H}_4$;

$\mathbf{P}_3^{(2,\mathrm{td})} \leftarrow \mathcal{R}\big(\mathrm{Cat}(\uparrow\mathbf{P}_4^{(2,\mathrm{td})},\ \tilde{\mathbf{P}}_3)\,;\ 1\big) + \gamma_3^{(2,\mathrm{td})}\cdot\mathcal{H}_3$;

$\hat{\mathbf{P}}_3 \leftarrow \mathbf{P}_3^{(2,\mathrm{td})}$; $\hat{\mathbf{P}}_4 \leftarrow \mathbf{P}_4^{(2,\mathrm{td})}$; $\hat{\mathbf{P}}_5 \leftarrow \tilde{\mathbf{P}}_5$;

return $\{\hat{\mathbf{P}}_3, \hat{\mathbf{P}}_4, \hat{\mathbf{P}}_5\}$

TD–BU sweep: the shallow levels ($\mathbf{P}_3$, $\mathbf{P}_4$) suffer most from insufficient semantic context, and a second top-down pass from $\mathbf{P}_5$ directly addresses this deficit, while a second bottom-up branch yields diminishing returns (Table VIII) since $\mathbf{P}_5$ already receives the richest semantic signals from Round 1.

Complexity analysis (YOLOv13n instantiation). Each Round 2 `DSC3k2` block follows the C2f split-and-merge architecture with minimal depth (n=1): 1×1 input/output projections, a single `DSC3k` sub-module containing two `DSBottleneck` layers of dual depthwise-separable convolutions, and one scalar FullPAD gate. For YOLOv13n with C_3=64 and C_4=128, the two Round 2 blocks contribute approximately 0.06M convolution parameters; together with channel-alignment layers and FullPAD gates, the total Round 2 overhead amounts to approximately 0.15M parameters and 0.7G FLOPs, representing $\sim$6% and 11% of the 2.48M / 6.4G baseline. On architectures without gated priors (e.g., YOLOv11n), the overhead is even smaller (0.10M, 3.8%) since only the compact refinement blocks are required.

C. Multi-Spectral Channel Attention Bottleneck (MSCA)

Beyond the macro-level topology addressed by MR-FPN, a second source of information loss lies in single-frequency spatial pooling within each fusion block. As discussed in Section II, Global Average Pooling retains only the DC component and discards higher-frequency texture and boundary cues critical for detection. Rather than proposing a new attention mechanism, our goal with MSCA is to investigate whether existing frequency-domain channel attention can be adapted to the nano regime at *negligible cost*—and, if so, to understand where and why it helps. We therefore adopt DCT-based multi-spectral channel attention from FcaNet [17], adapting it into YOLOv13's `DSC3k2` blocks. As we will

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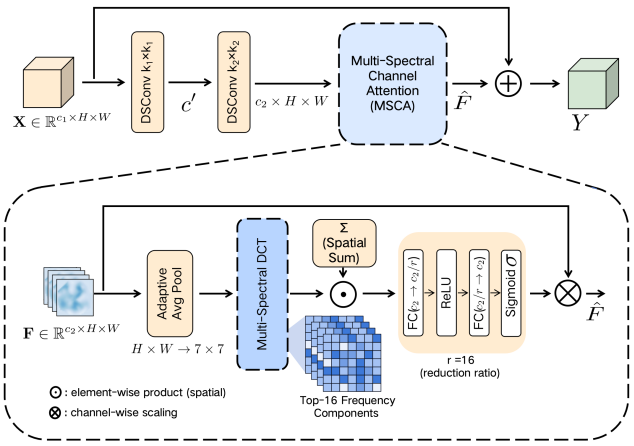


Fig. 4. Structure of the MSCA-Bottleneck. The block input $\mathbf{X}$ passes through two `DSConv` layers to produce $\mathbf{F}$, which is recalibrated by MSCA and added via a residual connection. Inside MSCA: features are pooled to a 7×7 grid, multiplied with Top-16 non-normalized DCT bases, spatially summed, and passed through a dual-FC bottleneck with r=16 to produce the attention vector $\mathbf{s}$.

demonstrate, the resulting module contributes +0.14 $AP_{50:95}$ while adding merely 0.01M parameters and zero FLOPs.

The core of MSCA is replacing the single-frequency GAP with a multi-spectral DCT descriptor. Given a 2D spatial position (p, q) in a feature map of size $S \times S$, the DCT-II basis function at frequency (u, v) is:

$$B^{u,v}(p,q) = \cos\Big(\frac{\pi\, u\,(p+0.5)}{S}\Big)\cos\Big(\frac{\pi\, v\,(q+0.5)}{S}\Big). \quad (6)$$

Following the mathematical formulation in [17], which omits the orthogonal normalization factors from the standard DCT-II, we observe that this choice is particularly effective in our detection context: the unnormalized basis amplifies the magnitude of high-frequency responses, pushing the subsequent sigmoid toward its saturation regions (≈0 or ≈1) and producing a *near-binary gating* effect that decisively separates foreground channels from background noise (validated in Table IX).

To compute the descriptor, a feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ is first resized to a canonical $S\times S$ (S=7) grid via adaptive average pooling, yielding $\tilde{\mathbf{F}} \in \mathbb{R}^{C\times S\times S}$. The C channels are evenly partitioned into K=16 groups, each assigned a distinct frequency pair (u_k, v_k) selected via the Two-Step criterion of [17]. The frequency-aware channel descriptor $\mathbf{d} \in \mathbb{R}^C$ is then:

$$d_c = \sum_{p=0}^{S-1}\sum_{q=0}^{S-1} \tilde{F}_{c,p,q} \cdot B^{u_{k(c)},\, v_{k(c)}}(p,q), \quad (7)$$

where $k(c) = \lfloor cK/C \rfloor$ maps channel c to its frequency group. This formulation compresses different channel groups with different DCT bases, encoding both low-frequency statistics and high-frequency texture cues in a single descriptor.

The descriptor $\mathbf{d}$ is passed through a dual-FC bottleneck to produce channel attention weights:

$$\mathbf{s} = \sigma\big(\mathbf{W}_2\, \delta(\mathbf{W}_1\, \mathbf{d})\big), \quad (8)$$

where $\mathbf{W}_1 \in \mathbb{R}^{(C/r)\times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C\times(C/r)}$ with fixed reduction ratio r=16, δ is ReLU, and σ is sigmoid. We fix r=16 across all layers regardless of channel width C.

Rank-1 attention as implicit regularization. In the shallowest backbone stage where C=16, $C/r = 1$ and the dual-FC bottleneck degenerates to a rank-1 mapping: $\mathbf{s} = \sigma(\mathbf{w}_2 \cdot \delta(\mathbf{w}_1^\top \mathbf{d}))$, where $\mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^C$ and the intermediate activation $\mathbf{w}_1^\top \mathbf{d} \in \mathbb{R}$ is a single scalar. All C channel scales are thus determined by one scalar, constraining the attention to learn a *global foreground-versus-background partition* rather than fine-grained per-channel modulation. With only $2C$ learnable parameters, this rank-1 bottleneck acts as a structural regularizer that prevents overfitting channel-wise statistics in the narrow-channel regime, explaining why the fixed r=16 consistently outperforms adaptive variants (Table IX). At wider layers where $C \geq 64$, the bottleneck dimension $C/r \geq 4$ recovers sufficient rank for meaningful channel-wise recalibration. The MSCA module is integrated into YOLOv13's `DSBottleneck` after the dual `DSConv` layers, yielding the complete forward pass (Fig. 4):

$$\mathbf{Y} = \mathbf{X} + \mathrm{MSCA}\big(\mathrm{DSConv}_2(\mathrm{DSConv}_1(\mathbf{X}))\big), \quad (9)$$

where $\mathrm{MSCA}(\mathbf{F}) = \mathbf{F} \otimes \mathbf{s}$ applies channel-wise scaling.

MSCA and MR-FPN enhance different granularities—intra-block channel recalibration versus inter-level cross-scale fusion—at different cost scales (0.01M versus 0.15M parameters). Their composition is investigated empirically in Section IV-C.

IV. Experiments

To validate the effectiveness and efficiency of our proposed YOLO-Prism, we conduct extensive experiments on three benchmarks. Section IV-A details the experimental settings. Section IV-B compares YOLO-Prism with state-of-the-art nano-scale detectors, including edge-hardware latency measurements. Section IV-C isolates the contributions of MR-FPN and MSCA and analyzes design choices, gradient-guided selective placement, cross-dataset gradient validation, and hyperparameter sensitivity. Section IV-D tests the cross-architecture transferability of MR-FPN to YOLOv11n. Section IV-E provides detection comparisons and feature response visualizations.

A. Experimental Setup

Datasets. We evaluate on three benchmarks and report standard COCO-style metrics: $AP_{50:95}$, AP_{50}, AP_{75}, and scale-specific AP_S, AP_M, AP_L where applicable. (i) *MS COCO 2017* [55]: 118K training and 5K validation images with 80 categories; all six metrics are reported on `val2017`. (ii) *PASCAL VOC 2007* [56]: 4952 test images spanning 20 categories, used exclusively for **cross-domain** evaluation. VOC's 20 classes form a subset of COCO's 80 classes with direct one-to-one mapping; models are evaluated directly with COCO-trained weights (no VOC fine-tuning), and predictions for non-VOC categories are ignored during evaluation. (iii) *VisDrone 2019* [57]: 6471 training and 548 validation images capturing 10 urban object categories from drone perspectives,

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TABLE II
COMPARISON WITH STATE-OF-THE-ART LIGHTWEIGHT DETECTORS ON COCO VAL2017. †: TRAINED FOR 200 EPOCHS UNDER OUR PROTOCOL; OTHERS USE OFFICIAL PRE-TRAINED WEIGHTS. THE YOLOV13N AND YOLO-PRISM RESULTS ARE THE MEANS OF THREE TRAINING SEEDS (7, 42, 123): 41.01, 41.05, 41.07 (STD 0.03) AND 41.64, 41.66, 41.68 (STD 0.02), RESPECTIVELY. "–" INDICATES METRICS NOT PUBLICLY REPORTED.

Model	**FLOPs** (G)	**Params** (M)	**Lat^{GPU}** (ms)	**Lat^{CPU}** (ms)	**FPS** (bs=32)	**AP$^{val}_{50:95}$**	**AP$^{val}_{50}$**	**AP$^{val}_{75}$**	**AP$^{val}_{S}$**	**AP$^{val}_{M}$**	**AP$^{val}_{L}$**
YOLOv6-3.0n [7]	11.4	4.70	–	–	–	37.50	–	–	–	–	–
Gold-YOLO-N [16]	12.1	5.60	–	–	–	39.90	55.94	–	19.15	44.32	57.75
YOLOv8n [58]	8.7	3.15	$13.70_{\pm 2.02}$	57.41	2827	37.35	52.59	40.61	18.57	41.04	53.47
YOLOv9t [8]	8.2	2.09	$35.42_{\pm 5.78}$	123.06	996	38.02	52.89	40.95	18.37	41.89	53.66
YOLOv10n [9]	8.6	2.76	$19.27_{\pm 3.13}$	82.00	1368	38.47	53.77	41.69	19.03	42.34	54.59
YOLOv11n [59]	6.5	2.62	$18.02_{\pm 2.76}$	70.70	1996	39.38	55.24	42.73	19.90	43.28	57.04
YOLOv12n [60]	6.5	2.59	$24.39_{\pm 4.00}$	91.68	1033	40.56	56.66	43.83	20.12	45.23	58.18
YOLOv13n† [11]	6.4	2.48	$35.97_{\pm 6.07}$	129.72	587	$41.04_{\pm 0.03}$	56.86	44.53	20.91	44.93	59.71
YOLO-Prism†	**7.1**	**2.63**	**41.23**$_{\pm 6.66}$	**148.64**	**560**	**41.66**$_{\pm 0.02}$	**57.88**	**45.02**	**21.21**	**46.19**	**59.54**

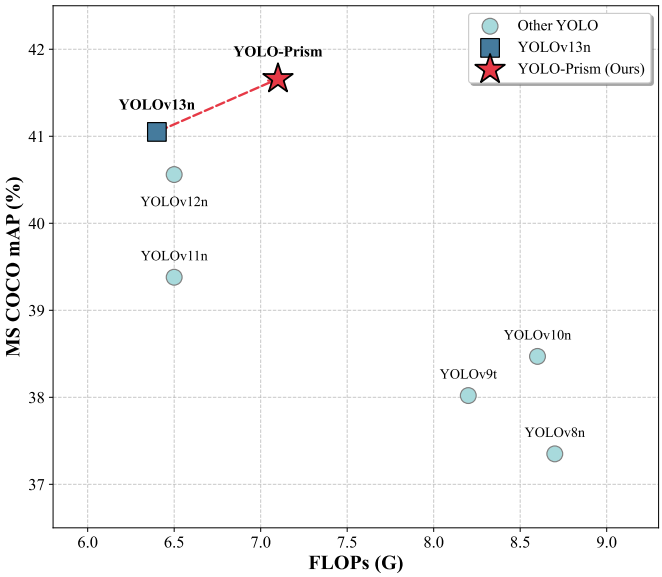


Fig. 5. MS COCO mAP versus FLOPs on COCO val2017. YOLO-Prism achieves the highest accuracy among all compared nano-scale detectors at competitive computational cost.

used to assess **transfer learning** by fine-tuning from COCO pre-trained weights.

Implementation Details. Our implementation is based on the Ultralytics framework (v8.3.63) with PyTorch 2.4.1. All models are trained at 640×640 resolution on a single RTX 4090 GPU. For the *COCO* benchmark and the main component ablation (Table VI), both YOLO-Prism and our YOLOv13n baseline are initialized from the official YOLOv13n pre-trained weights and trained for 200 epochs with a batch size of 56 under identical settings, using the default hyperparameters and data-augmentation pipeline. This ensures that any observed improvement is attributable solely to our architectural modifications rather than differences in training protocol, initialization, or data processing. The remaining YOLO-series detectors (YOLOv8n–YOLOv12n) are evaluated using their official pre-trained weights as released by their respective authors; Gold-YOLO-N [16] and YOLOv6-3.0n [7] results are directly cited from their original publications. For the *VisDrone* transfer experiment, models are fine-tuned for 200 epochs with a batch size of 24; the from-scratch control uses identical settings except for random initialization. The sub-component design ablations (Tables VIII, IX and XI) use a reduced 100-epoch, batch-24 setting.

Hardware and Speed Measurement. All latency, memory, and power measurements are conducted at batch = 1. GPU latency is measured under FP16 on synthetic inputs to isolate model inference from I/O and post-processing, using CUDA events for precise timing, averaged over 600 samples after 30-iteration warm-up, and reported as mean±std across three independent measurement runs. CPU latency is measured under FP32, averaged over 200 samples after 30-iteration warm-up. Peak GPU memory and inference power are also measured at batch = 1. Throughput (FPS$_{bs=32}$) is measured under FP16 with batch = 32 over 3200 images, reporting inference-only speed, consistent with Gold-YOLO [16] and YOLOv13 [11]. Multi-platform latency is additionally measured on RTX 2080 Ti, RTX 4090, and A100 GPUs alongside the Apple M1 CPU, under the same protocol. Model sizes are reported for the PyTorch format. We further verified that YOLO-Prism exports to ONNX (opset 18) without any architectural modification, and that its outputs are numerically consistent with the PyTorch implementation over representative COCO val2017 images (cosine similarity of 1.000 and a mean absolute difference below 4×10^{-6}), confirming compatibility with standard conversion pipelines and providing a clear path toward TensorRT/NPU backends.

B. Comparison with State-of-the-Art

We evaluate YOLO-Prism from two complementary perspectives: detection accuracy on three benchmarks—COCO val2017 (Table II), PASCAL VOC 2007 (Table IV), and VisDrone 2019 (Table V)—and deployment efficiency across multiple hardware platforms (Table III).

Overall accuracy. Table II reports the quantitative comparison on COCO val2017. Both YOLOv13n and YOLO-Prism are trained under identical settings (200 epochs, batch size 56, same data augmentation), ensuring that observed differences

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TABLE III
MULTI-PLATFORM DEPLOYMENT CHARACTERIZATION.

Model	Model Size (MB)	GPU Mem (MiB)	M1 CPU (ms)	2080Ti (ms)	RTX 4090 (ms)	A100 (ms)	NMS (ms)
YOLOv13n [11]	5.2	58	$183.58_{\pm 4.15}$	$22.59_{\pm 1.05}$	$35.97_{\pm 6.07}$	$23.14_{\pm 1.00}$	1–8
YOLO-Prism	**5.6**	**60**	$\mathbf{210.32}_{\pm 5.32}$	$\mathbf{25.92}_{\pm 0.92}$	$\mathbf{41.23}_{\pm 6.66}$	$\mathbf{27.05}_{\pm 2.33}$	**1–8**

TABLE IV
CROSS-DOMAIN EVALUATION ON PASCAL VOC 2007 TEST SET.

Model	$AP^{val}_{50:95}$	AP^{val}_{50}	AP^{val}_{75}
YOLOv8n [58]	60.27	80.36	66.99
YOLOv9t [8]	61.85	80.18	68.12
YOLOv10n [9]	61.90	80.98	68.24
YOLOv11n [59]	62.66	81.96	69.14
YOLOv12n [60]	64.43	83.01	71.02
YOLOv13†n [11]	65.26	83.43	71.65
YOLO-Prism†	**65.71**	**83.43**	**72.32**

TABLE V
TRANSFER LEARNING ON VISDRONE 2019. RESULTS ARE THE MEANS OF THREE TRAINING SEEDS (7, 42, 123). †: FINE-TUNED FROM COCO PRE-TRAINED WEIGHTS FOR 200 EPOCHS UNDER OUR PROTOCOL.

Model	$AP^{val}_{50:95}$	AP^{val}_{50}	AP^{val}_{75}	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}
YOLOv8n† [58]	$18.74_{\pm 0.08}$	32.64	18.39	9.75	28.86	45.36
YOLOv11n† [59]	$18.55_{\pm 0.15}$	32.05	18.36	9.48	28.61	45.02
YOLOv13n† [11]	$18.18_{\pm 0.08}$	31.71	17.80	9.36	28.38	41.22
YOLO-Prism†	$\mathbf{18.46}_{\pm 0.04}$	**32.09**	**17.85**	**9.74**	**28.53**	**42.21**

reflect architectural merit rather than training protocol variation. Notably, the official YOLOv13n checkpoint was trained for 600 epochs with batch size 256 and reports 41.6 AP in the original paper; therefore our 200-epoch baseline is intrinsically weaker, making comparisons with official-weight models conservative. YOLO-Prism achieves the best $AP_{50:95}$ of 41.66% among all compared detectors, surpassing the YOLOv13n baseline by 0.62 points while adding only 0.15M parameters (6%). This improvement is particularly meaningful given the baseline's strong starting point: at 41.04 $AP_{50:95}$, YOLOv13n already represents a highly optimized nano-scale detector, and Section IV-D shows that six alternative enhancement modules of comparable or larger cost all fail to improve it. The baseline and YOLO-Prism results are the means of three independent training runs (seeds 7, 42, 123) with standard deviations of ± 0.03 and ± 0.02, respectively; the 0.62-point gain exceeds $20\times$ the baseline standard deviation, and a paired t-test across the three seeds confirms its significance ($t = 92.5$, $p < 0.001$). YOLO-Prism also leads in AP_{50} (57.88%) and AP_{75} (45.02%), indicating consistent improvement at both lenient and strict IoU thresholds. Compared with YOLOv12n, YOLO-Prism improves $AP_{50:95}$ by 1.10 points. The advantages over earlier detectors are more pronounced: YOLO-Prism outperforms Gold-YOLO-N by 1.76 points with less than half the parameters and 41% fewer FLOPs, and exceeds YOLOv8n by 4.31 points with 17% fewer parameters. As visualized in Fig. 1 and Fig. 5, YOLO-Prism occupies the upper-left corner in both plots, reflecting a favorable accuracy–complexity trade-off.

Scale-wise analysis. By object scale, YOLO-Prism achieves the best AP_S (21.21%) and AP_M (46.19%), improving over the baseline by 0.30 and 1.26 points, respectively. The substantial AP_M gain reflects MR-FPN's second top-down round enriching intermediate pyramid levels with stronger semantic context. For large objects, the baseline retains a marginal 0.17-point advantage (59.71 vs. 59.54), which is smaller than typical inter-seed variance; YOLO-Prism's AP_L remains the second-highest among all compared nano detectors. This pattern arises from MR-FPN's asymmetric topology: Round 2 enriches $\mathbf{P}_3$ and $\mathbf{P}_4$, which are most capacity-starved at nano scale, while $\mathbf{P}_5$, already receiving the deepest semantic signals, passes through unchanged. In the zero-sum capacity regime of nano-scale detection, prioritizing the hardest scales is a principled trade-off.

Efficiency analysis. On the efficiency side, YOLO-Prism operates at 7.1G FLOPs and 2.63M parameters with a GPU latency of 41.23 ms and batch-32 throughput of 560 FPS on an RTX 4090. The latency overhead relative to the baseline ($\sim$14%) stems from the additional Round 2 top-down fusion blocks, kept small by the minimal per-node configuration (n=1) and the asymmetric topology that omits the costly bottom-up pass in Round 2. Notably, YOLO-Prism operates at substantially lower computational cost than Gold-YOLO-N (12.1G/5.60M), YOLOv6-3.0n (11.4G/4.70M), and YOLOv8n (8.7G/3.15M) while delivering higher accuracy than all of them.

Table III reports multi-platform latency and memory footprint across Apple M1, RTX 2080 Ti, RTX 4090, and A100. YOLO-Prism adds a consistent 14–17% latency overhead across all four platforms. Model size increases by 0.4 MB and peak GPU memory (RTX 4090) by 2 MiB, consistent with the 0.15M parameter increase. NMS overhead is independent of model architecture. Both models draw 126–127 W on RTX 4090 and 238 W on A100, confirming that the added modules have negligible energy impact.

Cross-domain evaluation on VOC 2007. To validate the generalization ability, we evaluate all YOLO-series models directly with their COCO-trained weights on the PASCAL VOC 2007 test set without any fine-tuning. As shown in Table IV, YOLO-Prism achieves the highest $AP_{50:95}$ of 65.71% and AP_{75} of 72.32%, surpassing the YOLOv13n baseline by 0.45 and 0.67 points, respectively. Notably, AP_{50} is identical for both models at 83.43%, while AP_{75}—which demands stricter localization—shows a clear gain. Because AP_{50} also requires correct classification, the unchanged AP_{50} rules out

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TABLE VI
ABLATION STUDY ON COCO VAL2017. †: RANDOM MSCA PLACEMENT (SEED=42). ‡: GRADIENT-GUIDED SELECTIVE MSCA PLACEMENT.

MR-FPN	MSCA	FLOPs (G)	Params (M)	$AP^{val}_{50:95}$	AP_S	AP_M	AP_L
×	×	6.4	2.48	41.04	20.91	44.93	59.71
×	✓	6.4	2.49	41.18	21.63	45.37	59.66
✓	×	7.1	2.63	41.62	21.57	46.20	59.38
✓	✓	7.1	2.64	41.48	21.28	46.21	58.77
✓	✓†	7.1	2.63	41.45	21.25	46.20	58.65
✓	✓‡	**7.1**	**2.63**	**41.66**	**21.21**	**46.19**	**59.54**

TABLE VII
PER-LAYER MSCA GRADIENT NORM ANALYSIS ON THE NAÏVE COMBINATION MODEL. VALUES ARE PER-PARAMETER MEAN L2-NORM ($\times 10^4$, AVERAGED OVER 50 BATCHES). THE SAME NORMALIZATION CONVENTION IS USED FOR BOTH COCO AND VOC.

Layer	Region	COCO	VOC
model.2.m.0.MSCA	Backbone	312.4	627.1
model.4.m.0.MSCA	Backbone	≈ 0	≈ 0
model.17.m.0.m.0.MSCA	Neck (R1)	≈ 0	≈ 0
model.17.m.0.m.1.MSCA	Neck (R1)	≈ 0	≈ 0
model.21.m.0.m.0.MSCA	Neck (R1)	≈ 0	≈ 0
model.21.m.0.m.1.MSCA	Neck (R1)	≈ 0	≈ 0
model.26.m.0.m.0.MSCA	Neck (R1)	≈ 0	≈ 0
model.26.m.0.m.1.MSCA	Neck (R1)	12.9	34.7
model.30.m.0.m.0.MSCA	Neck (R1)	≈ 0	≈ 0
model.30.m.0.m.1.MSCA	Neck (R1)	≈ 0	≈ 0
model.34.m.0.m.0.MSCA	Neck (R2)	8.4	33.2
model.34.m.0.m.1.MSCA	Neck (R2)	11.8	43.3
model.38.m.0.m.0.MSCA	Neck (R2)	1.8	4.9
model.38.m.0.m.1.MSCA	Neck (R2)	≈ 0	≈ 0

class mismatch as a confounding factor: if the gain were an artifact of COCO-to-VOC label remapping, both AP_{50} and AP_{75} would be affected. The improvement therefore reflects genuinely tighter bounding-box localization, consistent with MR-FPN's enrichment of shallow-level spatial detail. These results demonstrate the generalization ability of the multi-round fusion strategy to unseen domains.

Transfer learning on VisDrone 2019. Table V reports three-seed fine-tuning results together with strong lightweight baselines (YOLOv8n, YOLOv11n). Under transfer learning, YOLO-Prism improves over the YOLOv13n baseline by 0.28 $AP_{50:95}$ on average. The gain is modest in absolute terms, though it exceeds the baseline's inter-seed standard deviation (± 0.08) by a factor of 3.5, so it constitutes directional evidence; the same caveat applies to the accompanying gains of 0.38 AP_{50}, 0.05 AP_{75}, and 0.38 AP_S. Among these, the AP_S improvement is the most consistent across seeds, supporting a small-object benefit that is also visible qualitatively in Fig. 7. Two observations contextualize these numbers. First, the model ordering differs from COCO: YOLOv8n (18.74) and YOLOv11n (18.55) surpass YOLOv13n (18.18) despite being weaker on COCO. VisDrone instances are dominated by small objects [57], so performance is limited by feature-map spatial resolution rather than semantic fusion capacity: all four detectors fall within a narrow 9.36–9.75 AP_S band, whereas their COCO gap is far wider, since YOLOv13n's COCO advantage derives mostly from medium/large objects. In this resolution-limited regime the larger-parameter YOLOv8n gains a slight edge, and the YOLOv11n vs. YOLOv13n ordering reverses relative to COCO, underscoring that VisDrone's scale distribution is not representative of general-domain nano-detector performance. Second, the from-scratch runs drop both models by roughly 1.5–1.8 $AP_{50:95}$ (16.65 vs. 18.18 for the baseline; 16.69 vs. 18.46 for YOLO-Prism), confirming the importance of pre-training for this 6.5K-image dataset; even without pre-training, YOLO-Prism retains a small advantage (16.69 vs. 16.65), consistent with the transfer setting. Finally, AP_L is computed from only 2.8% of VisDrone val instances (1,068 of 38,759); its large inter-model spread (41.22–45.36) is dominated by noise and should not be over-interpreted.

C. Ablation Study

To validate each proposed module, we structure the ablation study along three axes: component contributions (Table VI), design choices (MR-FPN variants in Table VIII, MSCA formulations in Table IX, and gradient-guided vs. random placement in Table VII), and mechanism analysis (learned FullPAD gates in Table X).

Independent contributions. Rows 1–3 of Table VI show that both modules independently improve the baseline. MSCA alone provides a 0.14-point $AP_{50:95}$ gain and a 0.72-point AP_S improvement at essentially zero cost: 0.01M parameters (0.4% of baseline) and no increase in FLOPs. This yields a parameter efficiency roughly an order of magnitude higher than typical attention modules, demonstrating that multi-spectral channel recalibration recovers discriminative texture cues that single-frequency pooling discards, with negligible overhead. MR-FPN alone delivers a gain of 0.58 $AP_{50:95}$ and a pronounced 1.27-point AP_M boost, indicating that the second top-down refinement round effectively strengthens intermediate pyramid levels with richer semantic context. These gains require only 0.7 GFLOPs and 0.15M parameters, representing a 5.3× parameter efficiency advantage over a conventional BiFPN-style second round, which would duplicate full fusion blocks at an estimated cost of 0.8M parameters. Measured on an RTX 2080Ti under identical conditions, MR-FPN adds approximately 12% more training time (21.9 → 24.5 min/epoch), 3% more peak inference memory (60.1 → 62.1 MiB), and 15% more inference energy (1.57 → 1.81 mJ/image) than the baseline, whereas MSCA adds only about 2% training time (21.9 → 22.3 min/epoch) and negligible memory, with its

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TABLE VIII
MR-FPN ARCHITECTURE ABLATION ON COCO VAL2017.

TD-only	Min. config	FLOPs (G)	Params (M)	$AP^{val}_{50:95}$	AP^{val}_{50}
–	–	6.4	2.48	40.77	56.62
✓	×	7.5	2.69	41.25	57.10
×	✓	7.9	3.25	41.35	57.02
✓	✓	**7.1**	**2.63**	**41.20**	**57.16**

TABLE IX
MSCA DESIGN ABLATION ON COCO VAL2017.

Norm.	Bias	r	$AP^{val}_{50:95}$	AP^{val}_{50}
–	–	–	40.77	56.62
✓	✓	adaptive	40.69	56.35
×	×	16	**40.87**	**56.73**

TABLE X
LEARNED FULLPAD GATE VALUES γ AFTER TRAINING CONVERGENCE. ALL GATES ARE INITIALIZED TO 0; NON-ZERO VALUES INDICATE ACTIVE UTILIZATION OF THE PRECOMPUTED PRIOR $\mathcal{H}$.

Stage	Level	Layer	γ
Initial Distribution	$\mathbf{P}_4$	12	$+0.872$
Initial Distribution	$\mathbf{P}_3$	13	-0.006
Initial Distribution	$\mathbf{P}_5$	14	$+0.455$
Round 1 Top-Down	$\mathbf{P}_4$	18	-0.053
Round 1 Top-Down	$\mathbf{P}_3$	23	$+0.199$
Round 1 Bottom-Up	$\mathbf{P}_4$	27	$+0.075$
Round 1 Bottom-Up	$\mathbf{P}_5$	31	-0.158
Round 2 Top-Down	$\mathbf{P}_4$	35	-0.099
Round 2 Top-Down	$\mathbf{P}_3$	39	$+0.287$

all-layer inference-energy overhead ($\sim$24%) reduced to $\sim$8% under the selective placement adopted in YOLO-Prism.

Naïve combination degrades AP_L. When MSCA is naively combined with MR-FPN by inserting it into *all* `DSC3k2` blocks (Row 4), the overall $AP_{50:95}$ drops to 41.48, which is *lower* than MR-FPN alone at 41.62. Crucially, AP_L drops from 59.38 to 58.77, a loss of 0.61 points, while AP_M remains comparable at 46.21 versus 46.20. In the multi-round MR-FPN topology, features traverse both Round 1 and Round 2 fusion paths; when every bottleneck applies the r=16 channel squeeze, the cascaded information bottleneck excessively suppresses the low-frequency global features that large objects rely on. This observation raises a natural question: *is MSCA beneficial at every position, or only at specific layers?*

Gradient-guided selective placement. To answer this question, we perform a per-layer gradient norm analysis on the naïve combination model (Row 4 of Table VI), freezing all non-MSCA parameters and computing the L2-norm of gradients through each MSCA layer over 50 batches. Table VII reveals a pronounced sparsity pattern: the MSCA layers at model.2 (backbone), model.26, model.34, and model.38 (neck) exhibit measurable gradients, while the remaining 9 positions collapse to near-zero values. This sparsity is a *structural consequence* of the multi-round topology: positions receiving already-refined features from upstream rounds have negligible room for further recalibration. Repeating the identical analysis on the VOC 2007 validation set, under the same per-parameter normalization, yields the same sparsity structure. The Spearman rank correlation between the COCO and VOC gradient norms across all 14 layers is $\rho = 0.994$ ($p < 0.001$), confirming that the relative importance ordering of MSCA positions is quantitatively invariant across datasets—the selective placement is governed by the fusion topology itself, not by dataset-specific statistics.

Guided by this analysis, model.2 exhibits the strongest gradient signal, while the neck positions at model.26 and model.34 show moderate but measurable gradients. By contrast, model.38 is an order of magnitude weaker than the other active neck layers. We therefore adopt a half-MSCA configuration at Layer 26 (only m.1) and a full MSCA at Layer 34 (both m.0 and m.1), excluding model.2 and model.38. The backbone layer is excluded because its gradient vanishes in the standalone MSCA-only model, indicating it is an artifact of MR-FPN's enriched gradient flow rather than a genuinely beneficial site. This selective configuration—retaining only three of the original fourteen MSCA positions—is validated by ablation (Table VI, Row 6 vs. Row 5) as the optimal placement.

Random placement validation. To verify that the gradient-guided selection is not an artifact of random chance, we compare against a random MSCA placement with the same parameter budget (Row 5): three DSC3k2 layers (model.2, model.4, and model.30) are randomly selected to receive full MSCA. The random variant achieves 41.45 $AP_{50:95}$, which is 0.21 points *below* the gradient-guided variant (Row 6) and 0.03 points *below* the naive all-layer MSCA combination (Row 4). This gap confirms that random placement provides no benefit—and may even slightly hurt—whereas the gradient-identified positions are structurally meaningful.

This selective strategy yields YOLO-Prism (Row 6), achieving 41.66 $AP_{50:95}$ while recovering AP_L to 59.54 and using *fewer* parameters than the naïve combination (2.63M versus 2.64M), confirming that the pruned layers were pure overhead.

MR-FPN architecture ablation. Table VIII examines two key design choices in MR-FPN's Round 2 architecture. Compared with the single-round baseline (Row 1), all MR-FPN variants consistently improve $AP_{50:95}$ by 0.43–0.58 points, confirming the effectiveness of multi-round fusion. Among them, doubling the Round 2 block capacity (Row 2) yields only a marginal 0.05-point advantage over the adopted configuration while adding 0.06M parameters and 0.4G FLOPs, confirming that the minimal per-node configuration is sufficient for the lightweight refinement role of Round 2. Adding a full TD+BU sweep (Row 3) improves $AP_{50:95}$ by 0.15 points but at a disproportionate cost of 0.62M additional parameters, a 25% increase, and 0.8G FLOPs, yielding poor parameter efficiency. The adopted MR-FPN (Row 4) jointly applies both principles, achieving the best accuracy–efficiency trade-off at only 0.15M additional parameters and 0.7G FLOPs over the baseline.

MSCA design choices. Table IX examines whether

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TABLE XI
COMPARED WITH THE REPRESENTATIVE FEATURE ENHANCEMENT MODULES INTEGRATED WITHIN THE SAME YOLOV13N BASELINE AND THE FOUR BUDGET-MATCHED CONTROLS (ROWS 8–11) THAT EACH ADD APPROXIMATELY 0.15M PARAMETERS.

Model	Architecture	Additional complexity	FLOPs (G)	Params (M)	$AP^{val}_{50:95}$
YOLOv13n [11]	Single-round FPN-PAN with gated prior distribution (HyperACE + FullPAD)	—	6.4	2.48	40.77
+ BiFPN [14]	Bi-directional weighted fusion, stacked rounds	+0.92M params, +4.2G per round	10.56	3.40	38.44
+ Gold-YOLO GD neck [16]	Gather-and-distribute, single round	+1.02M params, +4.5G	10.87	3.50	37.86
+ Dynamic Head [43]	Unified attention on detection head	+0.40M params, +0.1G	6.46	2.88	37.01
+ AugFPN [61]	Residual pyramid enhancement, single round	+1.70M params, +6.5G	12.94	4.18	38.07
+ NAS-FPN [44]	NAS-discovered cross-scale topology	+1.38M params, +4.4G	10.82	3.86	37.12
+ Recursive FPN [15]	Recursive full-stack refinement	+0.95M params, +4.2G per round (reused)	10.64	3.43	39.01
+ 0.15M conv block	Single 3×3 conv on the P4 branch	+0.15M params, +0.4G	6.8	2.63	37.38
+ 0.15M extra FPN fusion	BU-only second-round fusion, no gating	+0.15M params, +0.2G	6.6	2.63	37.25
+ 0.15M residual	C2f residual refinement on P3/P4	+0.15M params, +0.7G	7.1	2.63	37.70
+ 0.15M attention	Channel attention before detection heads	+0.15M params, +0.1G	6.5	2.63	36.91
+ MR-FPN (ours)	**Asymmetric top-down second round with prior reuse and gating**	**+0.15M params, +0.7G per round (prior reuse)**	**7.1**	**2.63**	**41.20**

TABLE XII
CROSS-ARCHITECTURE GENERALIZATION OF MR-FPN ON COCO VAL2017. †: TRAINED UNDER OUR PROTOCOL.

Model	Params (M)	FLOPs (G)	$AP^{val}_{50:95}$	AP^{val}_{50}	AP^{val}_{75}
YOLOv11n† [59]	2.62	6.5	38.73	54.44	41.83
YOLOv11n† + MR-FPN	**2.72**	**6.7**	**39.23**	**54.74**	**42.57**

the original FcaNet formulation transfers effectively to YOLOv13n's narrow-channel regime. The augmented variant (Row 2) with DCT-II orthogonal normalization, FC bias, and an adaptive reduction ratio degrades $AP_{50:95}$ by 0.08 points relative to the no-attention baseline (Row 1). In contrast, the original formulation (Row 3, adopted as MSCA) with non-normalized DCT, bias-free FC, and fixed $r{=}16$ yields a 0.10-point gain. The non-normalized DCT amplifies high-frequency responses to produce near-binary sigmoid gating that effectively separates foreground from background, while the fixed extreme bottleneck acts as a structural regularizer in the narrow-channel regime.

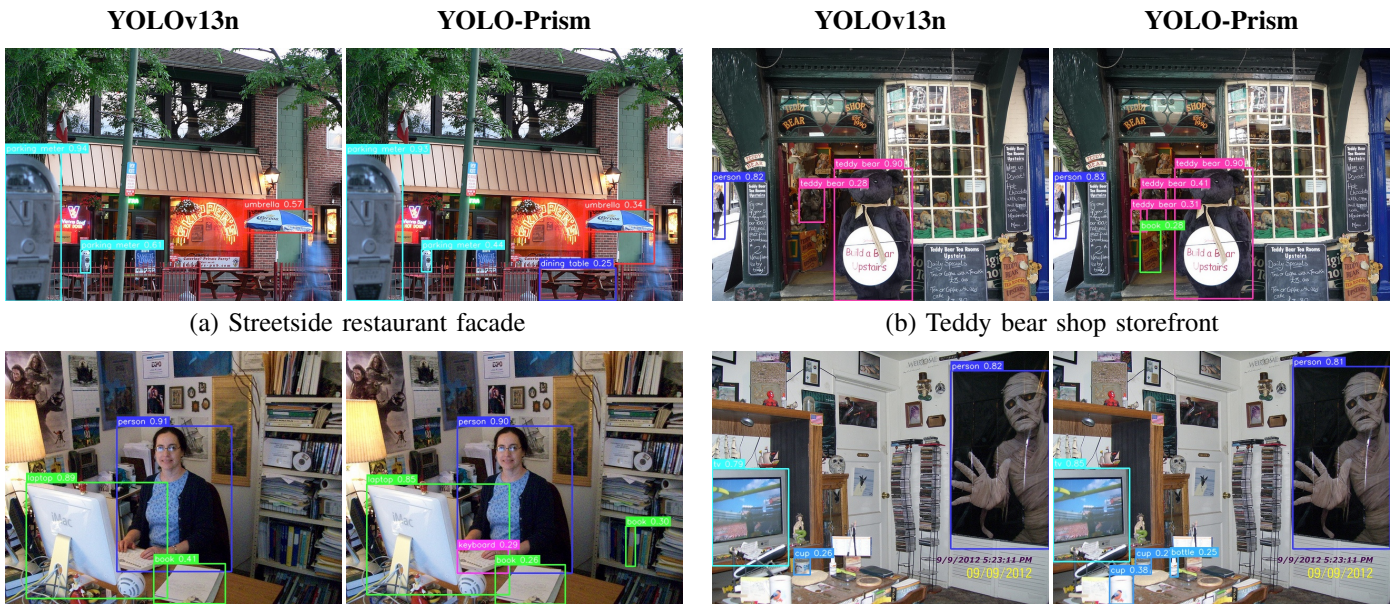
Learned Gate Dynamics. Table X reports the converged values of all nine FullPAD gates in the trained YOLO-Prism model. Two observations directly support our multi-round design. First, 8 of 9 gates deviate substantially from their zero initialization ($|\gamma|{>}0.01$), with 5 significantly active ($|\gamma|{>}0.15$), confirming that the model actively learns to gate $\mathcal{H}$ at each distribution point rather than bypassing the mechanism. Second, the shallowest level $\mathbf{P}_3$ exhibits progressive activation across rounds: $\gamma_{13}{=}{-}0.006$ (initial, near-dead), $\gamma_{23}{=}{+}0.199$ (Round 1 TD), and $\gamma_{39}{=}{+}0.287$ (Round 2 TD), increasing monotonically as more rounds of top-down semantic injection accumulate. By contrast, $\mathbf{P}_4$ saturates at the initial distribution stage ($\gamma_{12}{=}{+}0.872$) with its three subsequent gates all near zero ($|\gamma|{<}0.10$). This emergent sparsity—where the only gate to grow across rounds is $\mathbf{P}_3$'s—provides post-hoc evidence that MR-FPN's asymmetric topology correctly targets the pyramid levels most in need of additional refinement.

Sensitivity to hyperparameters and training configuration. Several observations from the ablation experiments collectively attest to the stability of our gains under hyperparameter and training-budget variation, without requiring additional experiments.

(i) **MR-FPN block budget**: Table VIII shows that varying the Round 2 configuration, from halving block capacity to expanding to a full bidirectional sweep, shifts AP by at most 0.15 points, while adding 25% more parameters for the full-sweep variant. The adopted minimal configuration sits at a clear optimum, but the performance surface is flat in its

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(a) Streetside restaurant facade

(b) Teddy bear shop storefront

(c) Cluttered office desk

(d) Indoor room with mixed objects

Fig. 6. Qualitative detection comparison on COCO val2017. In each pair, the left image shows YOLOv13n and the right shows YOLO-Prism. YOLO-Prism consistently detects additional objects missed by YOLOv13n while retaining all its detections.

vicinity.

(ii) **MSCA formulation**: the gain is robust to perturbations of the DCT configuration (Table IX), with the original fixed-r, bias-free formulation outperforming the augmented variant.

(iii) **Training budget**: the main ablation in Table VI uses 200 epochs and batch size 56, while the sub-component design ablations in Tables VIII and IX use 100 epochs and batch size 24. Despite halving the training budget, the gain magnitudes remain consistent, indicating that the improvements do not depend on a specific training schedule.

(iv) **Dataset, architecture, and hardware invariance**: gains persist across three datasets (Tables IV and V), across two backbone families (Table XII) and four hardware platforms (Table III), with the largest and most statistically reliable improvement on COCO and more modest, directionally consistent gains on the smaller-scale benchmarks. The MSCA gradient pattern is quantitatively consistent between COCO and VOC (Spearman $\rho = 0.994$, Table VII). This collective evidence indicates that our method's benefits are not confined to a single dataset, architecture, or hardware configuration.

D. Generalization and Comparative Analysis

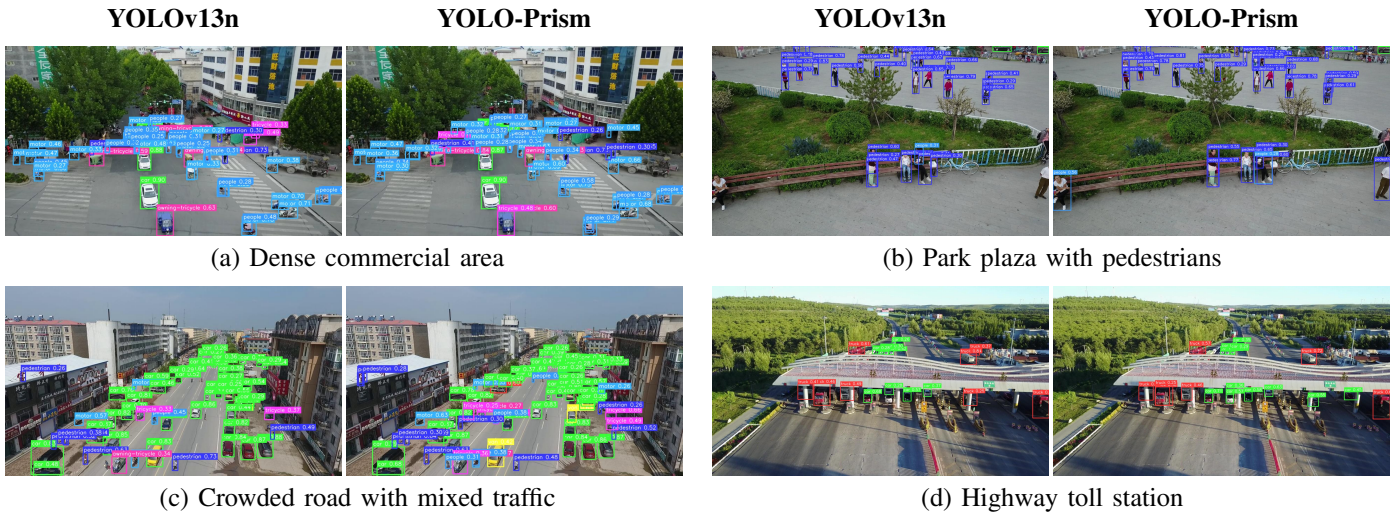
We further examine the broader utility of MR-FPN from two complementary perspectives. First, we compare MR-FPN against representative feature-enhancement modules integrated into the same YOLOv13n baseline (Table XI). Second, we investigate whether the asymmetric multi-round design transfers to detectors without gated priors, specifically YOLOv11n (Table XII).

Comparison with feature-enhancement modules. These results show that all six representative enhancement modules fail to surpass the baseline (37.01–39.01 vs. 40.77) despite substantially higher FLOPs (6.46–12.94 GFLOPs) and parameters (2.88–4.18M). To rule out insufficient parameter budget as the cause, we further construct four budget-matched controls (Table XI, rows 8–11) that each add about 0.15M parameters, the same budget as MR-FPN, covering a generic convolutional block, a bottom-up-only second-round fusion (the reverse direction of our Round 2), C2f residual refinement, and channel attention. They likewise fail to surpass the baseline (36.91–37.70) under the identical protocol. In contrast, MR-FPN improves the baseline to 41.20 with only 0.7 GFLOPs and 0.15M additional parameters, confirming that reusing the precomputed global correlation prior, instead of duplicating complete fusion modules or inserting generic blocks, is the key to efficient multi-round refinement at nano scale.

Cross-architecture generalization. A natural question is whether MR-FPN's effectiveness is intrinsically tied to YOLOv13's specific gated prior-distribution mechanism (HyperACE + FullPAD), or whether the asymmetric multi-round design offers broader utility. To test this, we append the same minimal Round 2 TD pass to YOLOv11n, whose neck employs a conventional C3k2-based FPN-PAN topology [13] without any global prior or learnable gating. Both models are trained under identical settings, and the results are shown in Table XII. MR-FPN improves YOLOv11n by 0.50 $AP_{50:95}$ at a cost of only 0.10M additional parameters and 0.2 GFLOPs. The gain magnitude is consistent with the 0.58 $AP_{50:95}$ observed on YOLOv13n (Table VI), despite the two baselines employing fundamentally different neck architectures. Notably, the parameter overhead on YOLOv11n is even smaller, 0.10M versus 0.15M, because its conventional C3k2 blocks are more compact than YOLOv13's DSC3k2 blocks. This result confirms that the benefit of a lightweight second TD pass is independent of YOLOv13's gating mechanism: nano-scale detectors across architectural families are systematically constrained by single-round cross-scale fusion, regard-

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(a) Dense commercial area (b) Park plaza with pedestrians

(c) Crowded road with mixed traffic (d) Highway toll station

Fig. 7. Detection comparison on VisDrone 2019 val set. YOLO-Prism recovers additional small objects in most scenes and sharpens detection confidence, though extremely dense scenes remain challenging. Consistent with the AP improvements in Table V.

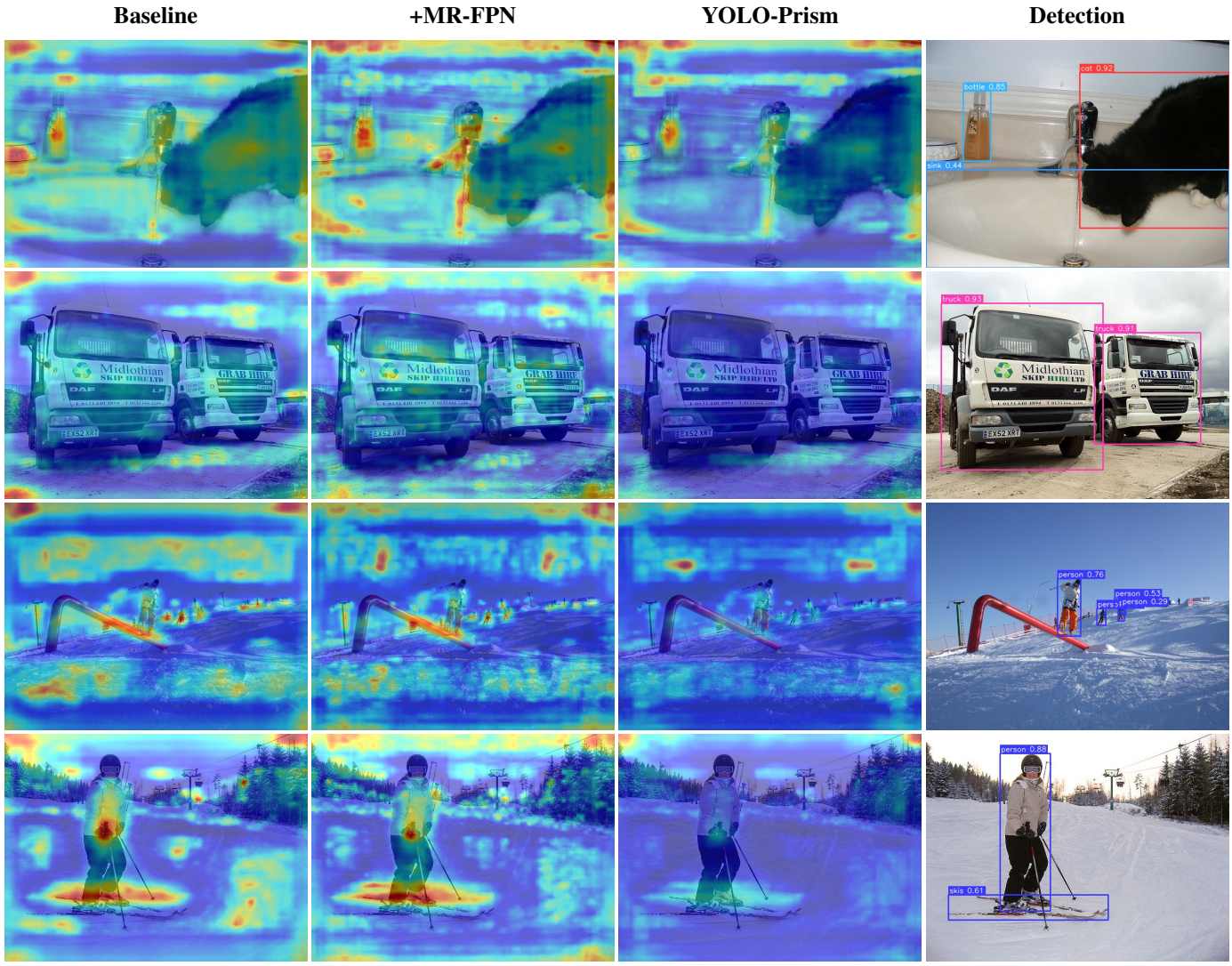


Fig. 8. Progressive background suppression across four COCO val2017 scenes. From left to right: Baseline, +MR-FPN, and YOLO-Prism detection-head response maps of $\mathbf{P}_3$–$\mathbf{P}_5$, followed by YOLO-Prism detections. Background responses are progressively suppressed, leaving cleaner foreground-focused features at each stage.

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less of whether the neck uses gated priors or conventional concatenation-based fusion.

Limitations and trade-offs. For a balanced assessment, we briefly discuss the limitations of MR-FPN and the scenarios where it is less effective. (i) *Extreme small-object regime.* On VisDrone, 68.6% of instances are smaller than 32×32 px and many occupy fewer than 10 px [57]; at the 640-pixel input scale such targets span only a few feature cells, so cross-scale fusion cannot recover the information lost to downsampling, and the VisDrone gain remains much smaller than on COCO. (ii) *Computational trade-offs.* MR-FPN adds 0.7 GFLOPs and 14–17% latency across platforms, and in extremely dense scenes the second top-down round can suppress ambiguous mid-confidence proposals, occasionally reducing detections at a fixed confidence threshold. (iii) *Scenarios.* The method is most beneficial for small- to medium-sized objects with adequate spatial resolution and less effective when targets are resolution-limited or when the baseline's single-round fusion is already saturated.

E. Qualitative Analysis

We complement the quantitative results with qualitative comparisons on COCO val2017 and VisDrone 2019, along with progressive feature response maps.

Detection comparison. Fig. 6 presents qualitative comparisons of YOLO-Prism with the YOLOv13n baseline on four representative COCO val2017 scenes. YOLO-Prism consistently detects more objects while retaining all baseline detections: a partially occluded dining table in (a), additional bears and a book sign amid cluttered merchandise in (b), a camouflaged keyboard and books in (c), and small cups among dense indoor clutter in (d). These examples demonstrate that MR-FPN's multi-round fusion effectively re-injects high-resolution spatial cues into deeper features, enabling the model to resolve occluded and camouflaged targets that single-pass fusion misses.

Fig. 7 extends the comparison to drone-captured VisDrone 2019 imagery, where most targets occupy only a few pixels. YOLO-Prism recovers additional objects in most scenes—notably +3 in the park plaza—though individual failure cases occur in extremely dense settings. For instance, in the dense commercial area (Fig. 7(a)), YOLO-Prism detects fewer objects than the baseline at the standard 0.25 confidence threshold (14 vs. 23), but this gap vanishes at lower thresholds (47 vs. 48 at 0.10), and reverses at higher thresholds (8 vs. 5 at 0.50). This confidence-dependent behavior reveals that MR-FPN's second top-down round suppresses ambiguous mid-confidence proposals—consistent with the reduced background interference observed in Fig. 8—while strengthening certainty on true positives. This pattern mirrors the quantitative results: MR-FPN's second top-down round injects deep semantic context into shallow features, which benefits medium-sized objects (AP_M +1.26 on COCO) but can suppress sub-10^2-pixel targets when semantic signals dominate over fine-grained spatial cues. The directionally consistent improvement on VisDrone (Table V) suggests that the multi-round fusion strategy may transfer to aerial surveillance, although the gain is statistically limited and extremely dense small-object scenes remain challenging.

Visualization: feature response comparison. Fig. 8 compares detection-head response maps of three configurations: Baseline, +MR-FPN, and YOLO-Prism. These maps, obtained by averaging the absolute channel-wise activations of $\mathbf{P}_3$–$\mathbf{P}_5$, highlight the regions the model attends to. Compared to the Baseline, which produces diffuse activations spread over both objects and background, MR-FPN generates substantially cleaner responses with reduced background interference, and YOLO-Prism further sharpens the foreground focus. This progressive improvement arises from two complementary mechanisms: MR-FPN's multi-round pathway allows each fusion stage to correct residual errors from the previous stage, achieving structural denoising, while MSCA's frequency-domain channel gating selectively amplifies task-relevant channels and suppresses uninformative activations. This two-stage suppression accounts for YOLO-Prism's perception advantage.

V. Conclusion and Future Directions

In this paper, we proposed YOLO-Prism, a nano-scale detector that establishes a new state of the art at only 2.63M parameters. Our primary contribution is the Multi-Round Feature Pyramid Network (MR-FPN), which introduces a "compute once, distribute many" paradigm. By exploiting the decoupling between the costly HyperACE computation and the lightweight FullPAD gates, MR-FPN reuses a single precomputed prior across a secondary top-down pass, substantially improving shallow-level representations at just 6% parameter overhead. Extensive experiments demonstrate that MR-FPN not only outperforms widely-used fusion modules (e.g., BiFPN, Gold-YOLO) under identical conditions but also provides consistent gains across different backbone families, improving YOLOv11n by 0.50 AP with only 3.8% overhead. Complemented by a near-zero-cost frequency-domain channel recalibration module, YOLO-Prism achieves 41.66% AP on COCO, surpassing all compared nano-scale detectors. Cross-dataset evaluations and multi-platform hardware measurements further support its robustness and deployment practicality.

We identify two directions for future work. First, the primary remaining limitation is the extreme small-object regime: on VisDrone 2019, many targets occupy fewer than 10 pixels, where even the enriched $\mathbf{P}_3$ features lack sufficient spatial resolution. Multi-scale inference, super-resolution pre-processing, or dedicated small-object head branches could extend the multi-round paradigm to such dense aerial scenes. Second, deployment on dedicated mobile NPUs (e.g., Snapdragon NPU) remains an open practical question, as it requires vendor-specific toolchains (QNN, SNPE) and quantization-aware optimization of the model.

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VI. Biography Section



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